

Spline

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1 Global weighted cubic fit with continuity constraints

1.1 Measured data

We start from a set of measured points

$$(x_j, y_j, \sigma_j), \quad j = 1, \dots, N_{\text{pts}}$$

where

- x_j is the coordinate,
- y_j is the measured value,
- σ_j is the measurement uncertainty.

We define weights

$$w_j = \frac{1}{\sigma_j^2}.$$

1.2 Spline parametrization

The domain is divided into N_{int} intervals centered on sample points $x_{c,i}$ with spacing dt .

Each interval i is described by a cubic polynomial

$$S_i(x) = a_i + b_i(x - x_{c,i}) + c_i(x - x_{c,i})^2 + d_i(x - x_{c,i})^3.$$

The vector of unknown parameters is

$$\theta = (a_0, b_0, c_0, d_0, a_1, b_1, c_1, d_1, \dots, a_{N_{\text{int}}-1}, b_{N_{\text{int}}-1}, c_{N_{\text{int}}-1}, d_{N_{\text{int}}-1})^T.$$

The total number of coefficients is

$$N_{\text{coef}} = 4N_{\text{int}}.$$

1.3 Weighted least-squares fit

For each measured point belonging to interval i we define

$$dx_{ij} = x_j - x_{c,i}.$$

The cubic prediction is

$$S_i(x_j) = a_i + b_i dx_{ij} + c_i dx_{ij}^2 + d_i dx_{ij}^3.$$

The weighted least-squares objective is

$$\chi^2 = \sum_j w_j (S_{i(j)}(x_j) - y_j)^2.$$

The normal equations of the least-squares problem are

$$A\theta = b$$

with

$$A = \sum_j w_j r_j r_j^T, \quad b = \sum_j w_j r_j y_j,$$

where r_j is the basis row vector associated with measurement j , containing zeros everywhere except for the four entries corresponding to the interval $i(j)$ to which the point belongs, where $r_j = (0, \dots, 0, 1, x_j - x_{c,i}, (x_j - x_{c,i})^2, (x_j - x_{c,i})^3, 0, \dots, 0)$.

1.4 Continuity constraints

At the boundary between intervals i and $i + 1$ we define the boundary point

$$x_b = x_{1,i}.$$

We define

$$dx_1 = x_b - x_{c,i}, \quad dx_2 = x_b - x_{c,i+1}.$$

1.4.1 Value continuity

We require the spline values to match:

$$S_i(x_b) = S_{i+1}(x_b).$$

Explicitly,

$$a_i + b_i dx_1 + c_i dx_1^2 + d_i dx_1^3 = a_{i+1} + b_{i+1} dx_2 + c_{i+1} dx_2^2 + d_{i+1} dx_2^3.$$

This produces the constraint

$$a_i + b_i dx_1 + c_i dx_1^2 + d_i dx_1^3 - a_{i+1} - b_{i+1} dx_2 - c_{i+1} dx_2^2 - d_{i+1} dx_2^3 = 0.$$

1.4.2 Derivative continuity

The derivative of the cubic polynomial is

$$S'_i(x) = b_i + 2c_i(x - x_{c,i}) + 3d_i(x - x_{c,i})^2.$$

At the boundary we impose

$$S'_i(x_b) = S'_{i+1}(x_b).$$

Explicitly,

$$b_i + 2c_i dx_1 + 3d_i dx_1^2 = b_{i+1} + 2c_{i+1} dx_2 + 3d_{i+1} dx_2^2.$$

Rearranging gives

$$b_i + 2c_i dx_1 + 3d_i dx_1^2 - b_{i+1} - 2c_{i+1} dx_2 - 3d_{i+1} dx_2^2 = 0.$$

1.5 Constraint matrix construction

Each continuity condition is a linear equation in the coefficient vector θ . These equations can be written in the form

$$C_k \theta = d_k,$$

where C_k is a row vector of length N_{coef} and d_k is the corresponding right-hand side (zero in our case).

In the implementation, each constraint row is constructed explicitly and stored in the container **Crows**. The vector **dvals** stores the corresponding right-hand side values.

For the value continuity constraint between intervals i and $i + 1$, the row vector has the structure:

$$C_k = (0, \dots, 0, 1, dx_1, dx_1^2, dx_1^3, -1, -dx_2, -dx_2^2, -dx_2^3, 0 \dots 0)$$

where the non-zero entries correspond to the coefficients (a_i, b_i, c_i, d_i) and $(a_{i+1}, b_{i+1}, c_{i+1}, d_{i+1})$.

Similarly, the derivative continuity constraint produces a row

$$C_{k+1} = 0, \dots, 0, 0, 1, 2dx_1, 3dx_1^2, 0, -1, -2dx_2, -3dx_2^2, 0, \dots, 0$$

Stacking all such rows produces the constraint matrix

$$C = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_{N_{\text{constr}}} \end{pmatrix},$$

with

$$N_{\text{constr}} = 2(N_{\text{int}} - 1).$$

The full set of constraints can therefore be written compactly as

$$C\theta = d,$$

where in the present case

$$d = 0.$$

1.6 Augmented least-squares system

The original weighted least-squares fit produces the normal equations

$$A\theta = b,$$

where

$$A = \sum_j w_j r_j r_j^T, \quad b = \sum_j w_j r_j y_j.$$

To enforce the linear constraints exactly, the problem is solved using Lagrange multipliers. A vector of multipliers

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_{N_{\text{constr}}})^T$$

is introduced and the system becomes

$$\begin{cases} A\theta + C^T\lambda = b, \\ C\theta = d. \end{cases}$$

This system can be written as the block matrix equation

$$\begin{pmatrix} A & C^T \\ C & 0 \end{pmatrix} \begin{pmatrix} \theta \\ \lambda \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix}.$$

The matrix

$$K = \begin{pmatrix} A & C^T \\ C & 0 \end{pmatrix}$$

is exactly the matrix constructed in the code.

The upper-left block A corresponds to the normal matrix of the weighted least-squares problem, while the lower-left block C contains the rows stored in **Crows**. The upper-right block C^T couples the Lagrange multipliers to the fit parameters.

Solving the augmented system yields simultaneously

- the spline coefficients θ ,

- the Lagrange multipliers λ enforcing the constraints.

Because the constraints appear explicitly in the block matrix, the solution satisfies

$$C\theta = d$$

exactly, ensuring both value and derivative continuity between neighbouring cubic segments.

1.7 Spline evaluation

Once the coefficients are obtained, the spline value in interval i is

$$S_i(x) = a_i + b_i(x - x_{c,i}) + c_i(x - x_{c,i})^2 + d_i(x - x_{c,i})^3.$$

This function is then used to reconstruct the continuous waveform.